

SAT UZBEKISTAN

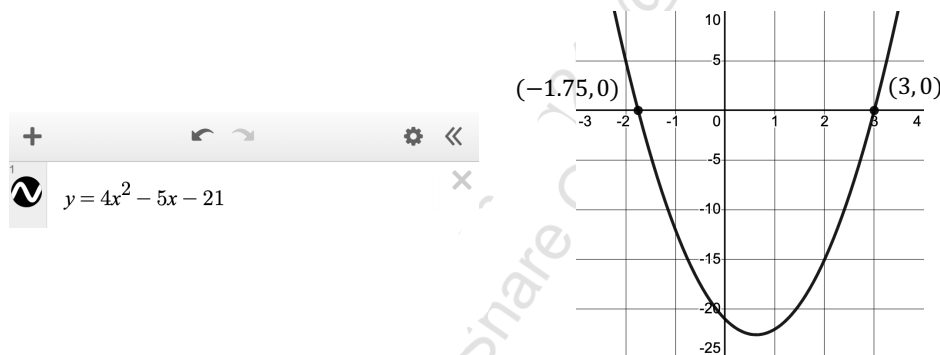
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Example: For $4x^2 - 5x - 21 = 0$, what is the sum of the solutions?

- A) $\frac{15}{4}$ B) $\frac{5}{4}$ C) $-\frac{5}{4}$ D) $-\frac{15}{4}$

Solution: To find the solutions (the x -intercepts), we write the function in $y =$ form and look for the x -intercepts. Here, we type the quadratic into Desmos as $y = 4x^2 - 5x - 21$.



The x -intercepts are at $(-1.75, 0)$ and $(3, 0)$. When a quadratic is equal to zero, as in this example, the solutions are the x -coordinate of the x -intercepts. So here, the solutions are $x = -1.75$ and $x = 3$. The sum of the solutions is $-1.75 + 3 = 1.25$. In fraction form, $1.25 = \frac{5}{4}$, so **the answer is B**.

This question can be solved algebraically by factoring the quadratic and setting each factor equal to zero, a method that we will learn in Chapter 11. However, for most students it is much easier to solve this question using this Desmos Hack!

Desmos Hacks Practice: Answers on page 326.

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1. If $42x - 225 = 18x + 309$, what is the value of x ?

- A) 3.5
B) 22.25
C) 309
D) 709.5

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3. If $22x + 549 = 6x + 717$, what is the value of x ?

- A) 10.5 B) 6 C) 4 D) 12

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2. What is the solution to the equation $5x^2 + 20x = -20$?

- A) -2
B) 0
C) 4
D) 10

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4. If $20.2x + 15.65 = 19 - 5.6x$, what is the value of x ?

- A) -0.25
B) 0
C) 0.13
D) 0.45

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5. For the equation $x^2 + 12x = 15 - 2x$, what is the smallest value of x that is a solution to the equation? (1)
A) -15 B) -12 C) -18 D) -21
6. Which of the following equations is equivalent to $2(x + 3)^2 - 4x$? (1)
A) $2x^2 + 2x + 9$
B) $2x^2 + 8x + 18$
C) $4x^2 - 4x + 18$
D) $4x^2 - 4x + 36$
7. For $x^2 + 4x - 396$, what is the sum of the solutions? (1)
A) -40
B) -22
C) -4
D) 4
8. For the system of equations below, the solution is at the point (a, b) . What is the value of b ? (1)
$$\begin{aligned} 80x - 47y &= 220 \\ 16x + 3y &= -80 \end{aligned}$$

A) -10 B) -12 C) -15 D) -18
9. For the equation below, what is the value of x ? (2)
$$\frac{5x}{6} - \frac{2}{3} = \frac{x}{6} + 2$$

A) 4 B) 6 C) 8 D) 10
10. If $5x + 8y = 67$ and $2x - y = 31$, what is the value of $x + y$? (2)
A) 16
B) 15.5
C) 15
D) 14
11. If one of the zeros to the equation $ax^2 - 24x - 64$ is at $x = 8$, what is the value of a ? (2)
A) 2
B) 4
C) 7
D) 10
12. What is the sum of the solutions to the equation $\frac{x^2}{4} - \frac{x}{8} - \frac{15}{8} = 0$? (2)
A) 1/2 B) 1/4 C) 1/3 D) 1/5
13. If x is a solution to the equation below, what is the value of $\sqrt{x + 10}$? (2)
$$\sqrt{x + 3} = x - 3$$

A) 4 B) 5 C) 6 D) 7
14. For the system of equations below, the solution is at the point (a, b) . What is the value of $3a - b$? (3)
$$\begin{aligned} 4x - y^2 &= 12 \\ 3x &= 9 \end{aligned}$$

A) 9 B) 12 C) 15 D) 18
15. For the system of equations below, one point is $(m, n - 13)$ where $m > 0$. What is the value of n ? (3)
$$\begin{aligned} y &= 0.25x^2 - 10x - 3 \\ 0.5x &= 17 - y \end{aligned}$$

A) 31
B) 18
C) 10
D) -3